

Inventory support
Fixed-asset support
Loan support
Customer correspondence
Project correspondence
Other approved document
8. Document Versioning and Preservation
Documents generated by the system must retain:

Document ID
Document type
Project ID
Customer ID
Master Sales Record version
Document version
Template version
Generation date and time
Generated by
File reference
Integrity hash
Status
Superseded-document reference
Signed documents must additionally retain:

Signature request reference
Signature audit certificate
Signer records
Signature completion dates
Final signed PDF reference
The system must:

Preserve original documents.
Preserve signed documents.
Preserve superseded documents.
Identify the current document version.
Prevent unauthorized replacement.
Prevent editing of a signed document.
Retain the document version used in each email or report.
Retain the exact attachment sent.
9. File Access Permissions
File access must be determined by:

Customer authorization
Project authorization
Document category
User role
Record status
Action type
Actions requiring permission checks include:

View
Upload
Replace

- Download
- Print
- Email
- Send for signature
- Archive
- Restore
- Delete or cancel
- Export
- Share internally

Signed documents and financial reports require stricter access controls than ordinary project notes.

Files must not be exposed through predictable or guessable storage paths.

10. File Upload and Integrity Controls

The system must validate:

- File type
- MIME type
- File size
- File name
- Malware or security scan result, where supported
- Checksum
- Source record
- User authorization

A failed or quarantined file must not be available as an official project document.

The system must preserve the original uploaded file and its integrity metadata.

Replacing a file requires a new file version or a new document version. It must not overwrite the original file.

11. Retention and Archival

The baseline requires permanent retention of:

- Original records
- Revisions
- Signed documents
- Signature audit certificates
- Commission reports
- Approved financial statements
- Audit records
- Relevant payment and payable evidence

The system must support:

- Retention classification
- Retention start date
- Retention status
- Archive status
- Legal or administrative hold indicator
- Authorized restoration
- Retention-policy version

Ordinary users must not delete permanently retained files.

Any future destruction process requires a separate approved retention policy and authorization workflow.

12. Sprint 14 Decisions

Decision 14-01 — Event-driven notifications

Notifications are created from controlled system events and workflow transitions.

Decision 14-02 — Notification channels

The initial notification channels are:

In-app notifications

Email

Decision 14-03 — Required notifications

Users may configure nonessential preferences but may not disable required workflow, security, approval, signature-expiration, or financial-report notifications.

Decision 14-04 — Template versioning

Email templates are versioned and the exact template version used for each message is retained.

Decision 14-05 — Attachment traceability

Every attachment must reference a stored document record and exact document version.

Decision 14-06 — Failed-delivery preservation

Failed emails remain preserved with their delivery status, failure reason, attachments, and retry history.

Decision 14-07 — Customer and project file structure

Each customer has a customer-level file area, and each project has a project-level file area linked to the customer.

Decision 14-08 — Document immutability

Signed documents, published financial reports, approved commission reports, and historical document versions cannot be silently replaced or edited.

Decision 14-09 — File replacement

A replacement creates a new file or document version. The original remains preserved.

Decision 14-10 — Access control

File view, download, print, email, signature, export, archive, and restore actions are permission-controlled and audited.

Decision 14-11 — Database authority

Files may be stored in an approved file-storage service, but the database remains authoritative for file metadata, relationships, versions, permissions, and integrity references.

Decision 14-12 — Permanent retention

Required historical documents, signed documents, approved financial reports, commission reports, and audit-supporting files are retained permanently.

Decision 14-13 — Annual contractor-report process

The system supports annual contractor summaries generated every January 5 and emailed to each applicable contractor.

Decision 14-14 — No duplicate signature workflows

Sprint 14 integrates with Sprint 08 and does not create a separate signature-request system.

Decision 14-15 — Notification auditability

Notification creation, delivery, failure, retry, read status, and preference changes are auditable.

13. Open Questions

Open Question 14-01 — File-storage provider

Which file-storage service will store uploaded and generated files?

Recommended default: Use an access-controlled object-storage service with encryption, versioning, checksums, and private access paths.

Open Question 14-02 — Maximum file size

What maximum file size should be permitted?

Recommended default: Establish separate limits for ordinary uploads, generated PDFs, signed documents, and supporting evidence.

Open Question 14-03 — Accepted file types

Which file types should be accepted for general project files?

Recommended default: PDF, PNG, JPEG, DOCX, XLSX, and CSV, subject to security scanning and role permissions.

Open Question 14-04 — Email sender identity

Which sender addresses should be used for:

Customer documents

Signature requests

Contractor reports

Financial reports

System alerts

Recommended default: Use separate configurable sender identities with reply handling defined for each category.

Open Question 14-05 — Email delivery provider

Which email delivery service will be used?

Recommended default: Use a transactional email service that supports delivery events, bounce handling, retries, templates, and message identifiers.

Open Question 14-06 — Customer email consent

What consent or authorization rules apply before emailing customers?

Recommended default: Permit only operational messages related to an active customer or project and retain the recipient address and delivery purpose.

Open Question 14-07 — Contractor email changes

Who may change a contractor's report-delivery email address?

Recommended default: Require Sales Manager, Comptroller, or authorized database-administrator action with audit logging and confirmation.

Open Question 14-08 — Notification aggregation
Should multiple notifications be combined into a digest?

Recommended default: Keep urgent and required actions immediate; permit daily aggregation only for nonurgent notifications.

Open Question 14-09 — Read receipts
Should email open tracking be enabled?

Recommended default: Track provider delivery and bounce events. Enable open tracking only if approved for the applicable recipient and communication type.

Open Question 14-10 — File retention policy
What retention classifications and legal holds are required beyond the baseline permanent-retention records?

Recommended default: Define classifications in Sprint 15 and prohibit deletion until the retention policy is approved.

Open Question 14-11 — File restoration
Who may restore archived files?

Recommended default: Sales Manager, Comptroller, and authorized Database Administrator users, with a required restoration reason.

Open Question 14-12 — Customer correspondence
Should ordinary emails be copied automatically into the customer/project file?

Recommended default: Store only system-generated and explicitly recorded operational correspondence initially. Defer mailbox synchronization.

Open Question 14-13 — Notification retry policy
How many times should failed emails be retried?

Recommended default: Use configurable exponential retry for transient errors and create a manual exception after the retry limit is reached.

Open Question 14-14 — External recipient verification
Should the system require confirmation before sending documents to a newly entered external email address?

Recommended default: Require authorized confirmation for sensitive documents, including contracts, invoices, certificates, financial reports, and commission reports.

Open Question 14-15 — Customer portal links
Should customers receive links to documents through a portal rather than attachments?

Recommended default: Defer a customer portal. Use controlled document links or attachments according to the approved delivery policy.

14. Acceptance Criteria

Sprint 14 is accepted when:

Controlled system events create the required notifications.
Notifications can be delivered in-app and by email.
Notification statuses distinguish created, queued, sent, delivered, failed, and read states.
Users can configure permitted notification preferences.
Required workflow and security notifications cannot be disabled.
Approval notifications link to the correct authorized workflow.
Signature notifications integrate with Sprint 08 without creating duplicate requests.
Payment-verification notifications link to the correct payment and project.
Commission reports are emailed only to applicable contractors.
Annual contractor summaries are generated and distributed through the approved process.
Failed email delivery creates an actionable exception.
Email retries do not create uncontrolled duplicate messages.
Each email preserves its template version and attachment references.
Customer and project file areas can be created and accessed by authorized users.
Generated, signed, approved, and superseded documents are retained.
Signed documents cannot be edited or silently replaced.
Every stored file has metadata, integrity information, and a record relationship.
Unauthorized users cannot view, download, print, email, or export files.
File replacement creates a new version and preserves the original.
Quarantined or failed files are not treated as official documents.
Project and customer file searches respect role permissions.
Required historical files are retained permanently.
File access, downloads, emails, and notification events are audit logged.
Failed notification and file-processing events appear on the appropriate dashboard.
Document attachments use the correct current or explicitly selected version.
Financial and contractor reports are sent only to authorized recipients.
Notification and file records remain linked to the applicable project, customer, document, report, or task.
Automated Sprint 14 tests pass.

15. Automated Test Scenarios

Notification Tests

Create a task assignment and confirm an in-app notification.
Create an approval task and confirm the authorized approver is notified.
Confirm an unauthorized user is not notified of restricted information.
Create a signature request and confirm integration with Sprint 08.
Create a payment-verification exception and confirm notification creation.
Mark a notification read and retain the read event.
Disable a permitted nonessential notification preference.
Attempt to disable a required notification and confirm rejection.
Confirm urgent notifications are not included only in a delayed digest.

Email Tests

Queue an email from an approved template.
Confirm the template version is retained.
Attach the correct document version.
Confirm delivery status changes are recorded.
Simulate a bounced email.
Confirm a failed message creates an exception.
Retry a transient failure.
Confirm retries do not create duplicate business events.

Confirm the exact sent attachment remains linked to the email record.

File Tests

Create a customer folder.

Create a project folder.

Upload a valid file.

Reject an unsupported file type.

Quarantine a failed security scan.

Store a generated contract.

Store a signed contract and signature certificate.

Supersede a document and confirm the original remains available.

Attempt to edit a signed document.

Attempt unauthorized download.

Archive and restore a file with proper authorization.

Confirm checksums or equivalent integrity values are retained.

Distribution Tests

Send an approved commission report to the applicable contractor.

Confirm another contractor cannot receive or view it.

Generate an annual contractor summary.

Confirm delivery failure creates an exception.

Send a financial report only to authorized recipients.

Confirm the distribution event is audited.

Security and Audit Tests

Attempt to access a project file without project authorization.

Attempt to use a predictable file path.

Attempt to attach an unauthorized document.

Attempt to send a document to an unauthorized recipient.

Confirm notification preference changes are audited.

Confirm file view, download, print, email, and archive actions are audited.

Verify the tamper-evident audit chain after notification and file activity.

16. Sprint 14 Completion Definition

Sprint 14 is complete when:

Notification events and delivery channels operate end to end.

Required workflow notifications are reliable and permission-controlled.

Customer and project files are organized and searchable.

Generated, signed, approved, and superseded documents are preserved.

Email messages and attachments are fully traceable.

Failed delivery and file-processing exceptions are actionable.

File access follows role, record, field, and action permissions.

Retention and archival metadata are implemented.

Notification and file activity is audit logged.

Acceptance tests pass.

Open questions are resolved or formally carried forward.

The Sprint 14 bundle is approved and preserved unchanged.

17. Next Markdown Bundle

The next bundle is:

text

GECC-15-testing-and-integrity.md

Its objective is to implement:

End-to-end testing
Unit and integration testing
Permission testing
Workflow-state testing
Audit-log testing
Document-consistency testing
Financial-reconciliation testing
Payment and payable testing
Inventory and asset testing
Commission testing
Notification and email testing
File-integrity testing
Backup and restore testing
Data-migration validation
Security testing
Performance testing
Failure-recovery testing
Regression testing
Test-data management
Release-blocking defects
Data-integrity monitoring
Audit-chain verification
Acceptance-test evidence
Production-readiness criteria
===== END FILE: GECC-14-Notification-And-Files.md =====

=== BEGIN FILE: GECC-15-Testing-And-Integrity.md ===

GECC-15-testing-and-integrity.md

GECC Sales Back Office System — Sprint 15

Bundle Metadata

Bundle name: GECC-15-testing-and-integrity.md

Project: GECC Sales Back Office System

Company: Go Ecco Climate Control

Sprint: 15

Sprint name: Testing, Security, and Data Integrity

Baseline: GECC-00-project-baseline.md

Previous bundle: GECC-14-notifications-and-files.md

Status: Implementation bundle

Date: 2026-09-20

Next planned bundle: GECC-16-deployment-and-operations.md

Primary database: Relational database

System of record: GECC database

Google Sheets: Export or reporting destination only

1. Sprint Objective

Implement the testing, security validation, data-integrity verification, backup testing, recovery testing, and release-readiness controls required before deployment.

Sprint 15 must ensure that:

Core workflows operate end to end.

Permissions prevent unauthorized access and changes.

Documents remain consistent with approved database records.

Financial calculations reconcile to source records.

Audit entries remain append-only and tamper-evident.

Backups can be restored successfully.
Notifications and files remain traceable.
Data migration can be validated.
Critical defects block release.
Production deployment has documented acceptance evidence.

2. Sprint Scope

2.1 Included

Test strategy
Unit testing
Integration testing
End-to-end testing
Regression testing
Workflow-state testing
Permission testing
Field-level security testing
Audit-log testing
Document-consistency testing
Financial reconciliation testing
Payment and payable testing
Inventory and asset testing
Commission testing
Notification and email testing
File-integrity testing
Backup and restore testing
Migration validation
Security testing
Performance testing
Failure-recovery testing
Test-data management
Defect management
Release-blocking criteria
Audit-chain verification
Data-integrity monitoring
Acceptance evidence

2.2 Excluded

Production deployment execution
Final production hosting configuration
Live customer-data migration
Penetration testing by an external provider, unless separately authorized
Formal financial-statement audit
Legal certification of electronic signatures
Disaster-recovery infrastructure procurement
Ongoing operations after deployment

3. Test Strategy

3.1 Test Levels

Testing must be performed at the following levels:

Unit testing
Component testing
Integration testing
Database constraint testing
Workflow testing
End-to-end testing

Security and permission testing

Performance testing

Backup and restore testing

User acceptance testing

Regression testing

3.2 Test Environments

The project must maintain separate environments for:

Development

Test or quality assurance

User acceptance testing

Production

Production data must not be copied into lower environments without approved de-identification or masking.

Each environment must have:

Environment identifier

Application version

Database version

Configuration version

Deployment date

Responsible administrator

Test-data source

Known limitations

3.3 Test Evidence

Each test execution must retain:

Test ID

Test name

Environment

Application version

Database version

Tester

Execution date and time

Input data

Expected result

Actual result

Pass or fail status

Evidence reference

Defect reference, if failed

Retest result

Test evidence must be retained according to the project's approved retention policy.

4. Test Data Management

4.1 Test Data Categories

Test data must include:

Customers with multiple contacts

Multiple project locations

Duplicate customer candidates

Projects in every lifecycle state

Projects with scope changes

Projects with multiple equipment records
Serialized and nonserialized inventory
Verified and unverified payments
Partial payments
Overpayments
Refunds and credits
Payables in every status
Positive-profit projects
Zero-profit projects
Negative-profit projects
Commission-eligible projects
Commission-adjustment projects
Fixed assets
Loans
Open and locked reporting periods
Signed and unsigned documents
Expired signature requests
Failed emails
Archived and superseded files
Unauthorized-access scenarios

4.2 Test Data Rules

Test data must:

Clearly identify its test purpose.
Avoid real customer secrets.
Avoid live payment information.
Avoid real bank references.
Avoid real contractor tax information.
Be resettable where required.
Support repeatable test execution.
Include known expected totals for reconciliation tests.

4.3 Production-Like Data Validation

Before migration, sample production-like records must be tested for:

Required fields
Duplicate records
Invalid project codes
Invalid statuses
Missing relationships
Invalid dates
Invalid amounts
Unsupported payment methods
Missing vendor references
Duplicate serial numbers

5. End-to-End Test Scenarios

5.1 Standard Project Lifecycle

Test the complete path:

Create customer.
Create project.
Generate project identification code.
Create Master Sales Record.
Approve Master Sales Record.

Generate invoice, contract, and certificate.
Complete contract signatures.
Complete invoice signatures.
Schedule installation.
Enforce the three-day waiting period.
Perform scope of work.
Complete Certificate signatures.
Create payables.
Record customer payment.
Verify payment.
Mark project complete.
Calculate commissions.
Include the project in the correct commission report.
Approve and distribute the report.
Include transactions in the appropriate financial period.
5.2 Scope-Change Lifecycle
Test:

Proposed scope change
Scope-change pending state
Authorized approval or rejection
New Master Sales Record version
Revised document generation
New signature sequence
Preservation of original records
Revised cost and profitability calculations
Commission impact
5.3 Payment Exception Lifecycle
Test:

Cashier's check entered without confirmation
Bank wire entered without reference or proof
Payment rejection
Payment verification
Partial payment
Overpayment
Refund
Payment reversal
Project-completion impact
Commission-adjustment impact
5.4 Financial-Period Lifecycle
Test:

Open period
Report preparation
Review
Sales Manager approval
Comptroller approval
Signature
Publication
Period lock
Formal adjustment
Preservation of original report

6. Permission and Security Testing

6.1 Access Dimensions

Testing must cover permissions at:

Screen level

Module level

Record level

Field level

Action level

Document level

Export level

Report level

API or service level, if applicable

6.2 Required Security Tests

The test suite must verify that:

Sales Associates see only authorized projects.

Sales Associates see only their own commission data.

Technicians see only assigned projects.

Contractors see only assigned records and fields.

Accounts Payable Associates cannot modify project costs.

Installation Managers cannot sign financial statements.

Database Administrators cannot approve financial records without assigned authority.

Unauthorized users cannot view payment amounts.

Unauthorized users cannot view profitability.

Unauthorized users cannot view payroll or commission reports.

Unauthorized users cannot access restricted files.

Unauthorized users cannot export restricted search results.

Permission failures are logged.

Search does not reveal restricted record existence.

Direct URL or object-reference manipulation cannot bypass authorization.

Archived and superseded files retain access controls.

6.3 Authentication Testing

Test:

Valid login

Invalid login

Failed-login logging

Logout

Session expiration

Concurrent sessions, if restricted

Password reset, if implemented

Disabled user access

Role changes

Access after role removal

Access after contractor status becomes inactive

7. Workflow and State-Machine Testing

Each state transition must be tested for:

Valid transition

Invalid transition

Required fields

Required approval

Required signature
Required payment verification
Required role
Required exception handling
Audit entry
Notification creation
Dashboard update
Specific tests must confirm that users cannot:

Skip Master Sales Record approval.
Generate documents from incomplete records.
Send an invoice before the contract is signed.
Schedule installation before required signatures.
Install before the three-day waiting period.
Complete work without required scope controls.
Complete a project before verified payment.
Earn commissions before project completion.
Publish financial reports without required approvals.
Edit locked periods without a formal adjustment.
Approve their own restricted action when separation of duties prohibits it.
8. Document-Consistency Testing
Generated documents must be compared against the approved source records.

Test consistency for:

Customer name
Customer contact details
Project location
Project identification code
Sales Associate
Scope of work
Equipment
Serial numbers
Retail price
Payment option
State-specific license artwork
Document version
Master Sales Record version
Signature fields
Revision references
The test suite must verify that:

Invoice, contract, and certificate data agree.
Revised documents use the revised Master Sales Record.
Original documents remain unchanged.
Superseded documents are clearly identified.
Signed PDFs remain linked to the correct signature audit certificate.
State-specific artwork is correct.
Unauthorized artwork overrides are blocked.
Generated documents cannot alter underlying project data.
9. Financial Reconciliation Testing
9.1 Ledger Tests
Verify:

text

Total Debits = Total Credits
Test:

Cash receipts
Customer refunds
Vendor payments
Payable creation
Inventory receipts
Inventory issues
Fixed-asset acquisition
Depreciation
Loan origination
Loan payments
Formal adjustments
9.2 Receivable Reconciliation
Verify:

text

Remaining Balance =
Adjusted Amount Due
- Verified Amount Paid
Reconcile:

Invoice totals
Approved adjustments
Verified payments
Refunds
Reversals
Accounts-receivable balances
9.3 Payable Reconciliation
Verify:

text

Outstanding Balance =
Amount Owed
- Amount Paid
Reconcile:

Certificate-triggered payables
Equipment payables
Dealer-fee payables
Materials payables
Vendor payments
Payable adjustments
9.4 Profitability Reconciliation

Reconcile:

Invoice amount
Equipment Cost
Material Cost
Dealer Fee
Lead Cost
Technician Cost
Damon Cost
Jerry Cost
Gross Profit
Ryan commission
Doug commission
Net Profit
Sales Associate share
GECC share
9.5 Financial-Report Reconciliation
Verify:

Balance sheet balances.
Ending cash reconciles to the ledger.
Accounts receivable agrees with source records.
Accounts payable agrees with source records.
Inventory agrees with FIFO layers.
Fixed assets agree with asset records.
Accumulated depreciation agrees with depreciation entries.
Loan balances agree with loan records.
Retained earnings reconcile.
Published report versions remain reproducible.
10. Audit-Log Testing
10.1 Audit Coverage
Verify audit logging for:

Login
Failed login
Logout
Views
Searches
Creates
Edits
Approvals
Exports
Prints
Downloads
Emails
Signatures
Permission failures
Restricted-access attempts
Administrative changes
Financial changes
File changes
Notification changes
Report publication

Period locking

Formal adjustments

10.2 Hash-Chain Testing

The test suite must verify that:

Each audit entry references the previous entry.

Each entry contains the expected integrity value.

Reordering an entry causes verification failure.

Deleting an entry causes verification failure.

Altering an entry causes verification failure.

Inserting an entry causes verification failure.

Replaying an entry is detected or rejected.

Verification results are recorded.

Only authorized users can run integrity verification.

Verification does not modify the audit chain.

10.3 Audit Immutability

Confirm that ordinary application users cannot:

Edit audit entries

Delete audit entries

Reorder audit entries

Disable audit logging

Bypass audit logging through exports or direct actions

11. Document, File, and Email Testing

Test:

File upload

File-type validation

File-size validation

Security scanning

File quarantine

File checksum

Document versioning

Superseded files

Signed-document preservation

File download authorization

File print authorization

Email attachment authorization

Email delivery

Email failure

Email retry

Duplicate-email prevention

Notification read status

Annual contractor-summary delivery

Financial-report distribution

File restoration

The exact attachment sent must remain associated with the email record.

12. Inventory, Asset, and Loan Testing

12.1 Inventory

Test:

Product identity

Serialized item
Nonserialized quantity
Receipt
Cost-layer creation
FIFO issuance
Multiple-layer issuance
Project allocation
Return
Damage
Scrap
Cost correction
Duplicate serial prevention
Negative inventory prevention

12.2 Fixed Assets

Test:

Vehicle
Computer
Office
Additional asset category
Acquisition
Placed-in-service date
Straight-line depreciation
Accumulated depreciation
Net book value
Disposal
Disposal adjustment
Historical depreciation preservation

12.3 Loans

Test:

Loan origination
Principal balance
Interest rate
Loan payment
Principal allocation
Interest allocation
Fee allocation
Loan correction
Loan payoff
Loan report reconciliation

13. Performance Testing

Performance testing must cover:

Login
Dashboard loading
Dashboard filtering
Project search
Customer search
Serial-number search
Financial report generation
Large document lists
Large audit-log searches

Commission-report generation
Annual-summary generation
File upload
File download
Email queue processing
Audit-chain verification
Database backup
Database restore
The project must define and record:

Expected response time
Maximum concurrent users
Maximum supported record volume
Maximum file size
Maximum report size
Maximum export size
Acceptable background-processing delay
Failure behavior when limits are exceeded

14. Backup and Restore Testing

14.1 Backup Requirements

Backups must include, as applicable:

Relational database
File metadata
Stored files
Configuration
Templates
Artwork versions
Audit records
Search-index rebuild information
Notification and email history

14.2 Restore Tests

Test restoration of:

Full database
Partial database
File metadata
Document files
Audit chain
Configuration
Template versions
Artwork versions
The restored system must verify:

Record counts
Foreign-key relationships
File references
File checksums
Audit-chain integrity
Ledger balances
Report reproducibility
Version-history preservation
14.3 Recovery Evidence

Each restore test must record:

- Backup identifier
- Backup date and time
- Restore date and time
- Restored environment
- Duration
- Records restored
- Files restored
- Integrity results
- Exceptions
- Tester
- Approval

15. Migration Validation

Before migration, the system must validate:

- Customer duplicates
- Project duplicates
- Invalid project codes
- Missing customer references
- Missing project locations
- Unsupported statuses
- Invalid dates
- Invalid currency amounts
- Duplicate serial numbers
- Missing vendor references
- Invalid contractor assignments
- Missing historical document references
- Inconsistent spreadsheet values

Migration validation must produce:

- Accepted-record count
- Rejected-record count
- Corrected-record count
- Duplicate count
- Unresolved-exception count
- Migration report
- Approval record

No production migration may proceed with unresolved release-blocking data errors.

16. Defect Management

16.1 Defect Severity

Defects must be classified as:

` BLOCKER

CRITICAL

HIGH

MEDIUM

LOW

16.2 Release-Blocking Defects

A release is blocked by:

Unauthorized access to restricted information

Unauthorized financial modification
Incorrect payment verification
Incorrect project completion
Incorrect commission calculation
Incorrect financial report totals
Unbalanced journal entry
Audit-log integrity failure
Data loss
Document inconsistency affecting legal or financial terms
Failed backup restoration
Duplicate or missing critical records
Inability to preserve signed or published documents
16.3 Defect Lifecycle
Permitted defect statuses:

OPEN
TRIAGED
IN_PROGRESS
RESOLVED
READY_FOR_RETEST
VERIFIED
REOPENED
DEFERRED
CLOSED

Deferred defects require:

Business impact
Risk assessment
Owner
Target sprint or release
Approval

17. User Acceptance Testing

User acceptance testing must include representatives or approved test owners for:

Sales Associate
Sales Manager
Comptroller
Installation Manager
Technician
Accounts Payable Associate
Database Administrator
Other contractor access, where applicable
User acceptance testing must confirm:

Workflow usability
Dashboard usefulness
Search accuracy
Document consistency
Payment and payable workflows
Commission reports
Financial reports
Notification handling
File access

Permission behavior

Error-message clarity

Each user acceptance test must have an owner and approval result.

18. Sprint 15 Decisions

Decision 15-01 — Multi-level testing

The project will use unit, integration, end-to-end, security, performance, backup, restore, migration, and user acceptance testing.

Decision 15-02 — Separate environments

Development, test, user acceptance, and production environments remain separate.

Decision 15-03 — Protected test data

Test environments use synthetic, masked, or otherwise approved data. Live sensitive records are not copied into test environments without authorization.

Decision 15-04 — Release-blocking security defects

Any defect that permits unauthorized access to restricted records, fields, documents, reports, or exports blocks release.

Decision 15-05 — Release-blocking financial defects

Any defect that causes incorrect payment status, project completion, profitability, commission, ledger balance, or financial-report totals blocks release.

Decision 15-06 — Audit integrity gate

The release is blocked if audit-chain verification fails or if required events bypass the audit system.

Decision 15-07 — Immutable historical records

Testing must verify that approved, signed, published, paid, posted, and audited records cannot be silently overwritten.

Decision 15-08 — Reconciliation gate

Financial reports cannot be approved for release unless required reconciliation tests pass.

Decision 15-09 — Backup and restore gate

A tested backup that cannot be restored with verified integrity is not acceptable for production release.

Decision 15-10 — Repeatable tests

Critical tests must be automated or repeatable using controlled test data.

Decision 15-11 — Defect evidence

Every failed test must reference a defect or approved explanation.

Decision 15-12 — UAT approval

User acceptance testing requires documented approval from authorized business representatives.

Decision 15-13 — Migration gate

Production migration requires a completed migration-validation report and approval of unresolved exceptions.

Decision 15-14 — Security test coverage

Security testing includes direct user-interface actions, search, exports, document links, APIs or services where applicable, and manipulated object references.

Decision 15-15 — Regression testing

Every release candidate must pass regression tests for all previously approved sprints.

19. Open Questions

Open Question 15-01 — External penetration testing

Is an independent penetration test required before production deployment?

Recommended default: Obtain an independent security review before production if budget and schedule permit. At minimum, complete documented internal authorization and access testing.

Open Question 15-02 — Performance targets

What exact response-time and concurrency targets define acceptable performance?

Recommended default: Establish targets by function before final release testing, including dashboard, search, report generation, file operations, and login.

Open Question 15-03 — Recovery objectives

What are the required:

Recovery Point Objective

Recovery Time Objective

Recommended default: Define these before Sprint 16 deployment planning and test restoration against them.

Open Question 15-04 — Backup frequency

How frequently must backups run?

Recommended default: Use scheduled daily full or equivalent backups with more frequent transaction or incremental protection where supported.

Open Question 15-05 — Backup retention

How many backup versions must be retained?

Recommended default: Define a rotation that includes recent daily backups, weekly backups, and monthly archival backups.

Open Question 15-06 — Test-data reset

Should test environments reset automatically after each test cycle?

Recommended default: Maintain repeatable seed data and provide an authorized reset process rather than automatically resetting shared environments.

Open Question 15-07 — Audit-chain verification frequency

How frequently should production audit-chain verification run?

Recommended default: Run scheduled verification at least daily and after any restoration or migration.

Open Question 15-08 — Migration reconciliation tolerance

What differences are acceptable between source and migrated data?

Recommended default: No unexplained differences in financial records, project codes, signed documents, payment records, or audit history. Any other tolerance requires explicit approval.

Open Question 15-09 — Data masking standard
Which fields require masking in lower environments?

Recommended default: Mask customer contact details, payment references, contractor contact information, financial values, signature information, and uploaded documents unless synthetic replacements are used.

Open Question 15-10 — UAT signers
Which individuals have authority to approve user acceptance testing?

Recommended default: Require approval from the project owner or Comptroller plus role representatives for operational areas affected by the release.

Open Question 15-11 — Accessibility testing
Which accessibility standard should be used?

Recommended default: Target WCAG 2.1 AA or the applicable current organizational standard.

Open Question 15-12 — Browser and device support
Which browsers and devices must be supported?

Recommended default: Define a supported browser matrix before production deployment and test the primary desktop workflows.

Open Question 15-13 — Defect acceptance
Who may approve deferral of a High-severity defect?

Recommended default: Require approval from the project owner and Comptroller, with documented risk and mitigation.

Open Question 15-14 — Disaster recovery environment
Will restoration occur in a separate recovery environment or the primary environment?

Recommended default: Use a separate recovery environment for testing to avoid destructive restoration tests.

20. Acceptance Criteria
Sprint 15 is accepted when:

Unit tests cover critical calculations and validation rules.
Integration tests cover cross-module data flow.
End-to-end tests cover the complete project lifecycle.
Scope-change workflows are tested.
Payment, verification, refund, and adjustment workflows are tested.
Payable creation and payment workflows are tested.
Inventory FIFO and serial-number rules are tested.
Fixed-asset depreciation and disposal are tested.
Loan principal and interest tracking are tested.

Cost, profitability, and commission calculations reconcile.
Financial reports reconcile to the ledger and source records.
Balance-sheet and cash-flow integrity tests pass.
Workflow-state bypass attempts are blocked.
Role, record, field, action, document, and export permissions are tested.
Restricted search and direct-record access attempts are blocked.
Unauthorized access attempts are audited.
Generated documents match approved Master Sales Record data.
Signed and published documents remain immutable.
Email attachments match the intended document version.
Notification delivery failures are handled and audited.
File upload, quarantine, versioning, archival, and restoration are tested.
Audit-chain tampering tests detect alteration, deletion, insertion, and reordering.
Posted journal entries cannot be edited or deleted.
Locked financial periods cannot be ordinarily edited.
Formal adjustments preserve original reports.
Backup restoration succeeds with verified database and file integrity.
Migration-validation results are documented.
No unresolved Blocker or Critical defects remain.
All deferred High-severity defects have documented approval.
Required user acceptance tests are approved.
Regression tests for Sprints 00–14 pass.
Performance results meet approved targets.
Release evidence is complete and reproducible.

21. Automated Test Scenarios

Core Regression Tests

Run the approved baseline acceptance tests.
Run Sprint 08 electronic-signature tests.
Run Sprint 09 payment and payable tests.
Run Sprint 10 cost and commission tests.
Run Sprint 11 inventory, asset, and loan tests.
Run Sprint 12 financial-reporting tests.
Run Sprint 13 dashboard and search tests.
Run Sprint 14 notification and file tests.

Integrity Tests

Alter a posted journal entry.
Delete an audit entry.
Reorder audit entries.
Insert an audit entry.
Alter a signed document.
Alter a published financial report.
Change an approved Master Sales Record without revision.
Change a historical commission calculation.
Change a closed-period transaction.
Confirm every attempt is blocked and audited.

Recovery Tests

Restore a full database backup.
Restore file metadata and stored documents.
Verify document checksums.
Verify foreign keys.
Verify ledger balances.
Verify audit-chain integrity.
Rebuild or restore the search index.

Reproduce a published financial report.
Confirm restored permissions and configurations.
22. Sprint 15 Completion Definition
Sprint 15 is complete when:

All critical workflows have automated or repeatable tests.
Security and permission controls are verified.
Documents, files, reports, and financial records reconcile.
Audit-chain integrity is verified.
Backup and restore testing succeeds.
Migration validation is documented.
Performance testing meets approved targets.
User acceptance testing is approved.
No unresolved release-blocking defects remain.
Deferred defects have documented owners and approvals.
Regression testing for Sprints 00–14 passes.
Release evidence is complete.
Open questions are resolved or formally carried forward.
The Sprint 15 bundle is approved and preserved unchanged.

23. Next Markdown Bundle

The next bundle is:

text

GECC-16-deployment-and-operations.md

Its objective is to implement:

Deployment instructions
Environment configuration
Production setup
Initial-user setup
Role assignment
Permission configuration
Data-migration plan
Migration execution controls
Backup procedures
Recovery procedures
Monitoring
Alerting
Audit-chain verification operations
Administrator guide
User guides
Operational runbooks
Release checklist
Rollback procedures
Incident response
Support procedures
Maintenance procedures
Configuration management
Version management
Production acceptance
Go-live readiness

Post-deployment validation

Handover and operational ownership

===== END FILE: GECC-15-Testing-And-Integrity.md =====

=== BEGIN FILE: GECC-16-Deployment-And-Operations.md ===

GECC-16-deployment-and-operations.md

GECC Sales Back Office System — Sprint 16

Bundle Metadata

Bundle name: GECC-16-deployment-and-operations.md

Project: GECC Sales Back Office System

Company: Go Ecco Climate Control

Sprint: 16

Sprint name: Deployment, Migration, and Operations

Baseline: GECC-00-project-baseline.md

Previous bundle: GECC-15-testing-and-integrity.md

Status: Implementation and go-live bundle

Date: 2026-09-20

Primary database: Relational database

System of record: GECC database

Google Sheets: Export or reporting destination only

1. Sprint Objective

Deploy the GECC Sales Back Office System into production, migrate approved existing data, configure initial users and permissions, establish backup and recovery procedures, and transfer the system into operational ownership.

Sprint 16 must ensure that:

Production is configured consistently with approved requirements.

Initial users receive correct roles and permissions.

Existing customer and project data is migrated safely.

Historical documents and records are preserved.

Backup and restore procedures are operational.

Monitoring and alerting are active.

Rollback procedures are documented.

Administrators and users have operating instructions.

Go-live approval is based on objective acceptance evidence.

Post-deployment validation confirms system integrity.

2. Sprint Scope

2.1 Included

Production deployment

Environment configuration

Database configuration

File-storage configuration

Email configuration

Electronic-signature configuration

Artwork and template configuration

Initial-user setup

Role assignment

Permission configuration

Data-migration planning

Data-migration execution

Migration reconciliation

Backup configuration

Restore procedures

Monitoring
Alerting
Audit-chain verification operations
Administrator guide
User guides
Operational runbooks
Release checklist
Rollback plan
Incident-response process
Support process
Maintenance process
Configuration management
Version management
Production acceptance
Go-live validation
Operational handover
2.2 Excluded
New business requirements
Major feature development
New payment methods
Automatic bank integration
General HR functionality
Technician GPS tracking
Technician time tracking
Customer self-service portal
Tax-return preparation
Unapproved accounting-policy changes
3. Production Architecture
The production environment must include:

Application services
Relational database
File-storage service
Email-delivery service
Electronic-signature integration
Background-task or job-processing service
Audit-log storage
Monitoring and alerting
Backup storage
Recovery environment or documented recovery destination
Search index or search service, if used
The production system must use approved configuration values for:

Company identity
Time zone
Email sender identities
Signature settings
Document templates
GECC logo
State license artwork
Financial-report periods
Commission rates
Default cost values

- File-size limits
- Notification rules
- Retention classifications
- User roles
- Backup schedules
- Secrets, credentials, signing keys, and service tokens must be stored in an approved secrets-management mechanism and must not be stored in source code or ordinary configuration files.

4. Deployment Deliverables

4.1 Deployment Package

The deployment package must include:

- Application release version
- Database migration scripts
- Configuration templates
- Environment-variable reference
- File-storage configuration
- Email configuration
- Signature-service configuration
- Search-index configuration
- Background-job configuration
- Backup configuration
- Monitoring configuration
- Rollback scripts or procedures
- Release notes
- Known limitations
- Dependency versions

4.2 Deployment Runbook

The deployment runbook must document:

- Pre-deployment backup
- Environment validation
- Release artifact verification
- Database migration
- Application deployment
- File-storage verification
- Email verification
- Signature-service verification
- Background-job verification
- Search-index verification
- Smoke testing
- Monitoring validation
- Go-live decision
- Rollback process

4.3 Configuration Register

The system must maintain a configuration register containing:

- Configuration item
- Current value or protected reference
- Environment
- Effective date
- Configuring user
- Approval status

Previous value reference
Change reason
Audit reference
Sensitive values must not be displayed in clear text.

5. Initial User Setup

5.1 User Records

Each initial user must have:

User ID
Name
Email
Phone, if required
Active status
Assigned role
Contractor or employee classification
Effective date
Supervisor or approval relationship, if applicable
Authentication status
Last-login information

5.2 Initial Roles

The production system must support the approved roles:

Sales Associate
Sales Manager
Comptroller
Installation Manager
Technician
Accounts Payable Associate
Database Administrator
Other Contractor

5.3 Role Assignment

Initial role assignments require:

Authorized requestor
Assigned role
Effective date
Approval, where required
Reason
Audit record

Users must receive only the minimum access required for their duties.

5.4 Initial-User Verification

Before go-live, verify:

Each user can authenticate.
Each user has the correct role.
Each user can access required screens.
Each user cannot access restricted screens.
Each user can perform required permitted actions.
Each restricted action is blocked and audited.

6. Data-Migration Plan

6.1 Migration Sources

Potential migration sources include:

- Existing Google Sheets
- Existing customer records
- Existing project records
- Existing project codes
- Existing documents
- Existing contractor records
- Existing vendor records
- Existing payment records
- Existing payable records
- Existing inventory records
- Existing fixed-asset records
- Existing loan records

Google Sheets remain migration sources or reporting outputs and do not become the system of record.

6.2 Migration Stages

Migration must use the following stages:

- Source inventory
- Source backup
- Source profiling
- Field mapping
- Duplicate detection
- Data cleansing
- Test migration
- Test reconciliation
- Business review
- Approved production migration
- Post-migration validation
- Migration sign-off

6.3 Migration Mapping

Each migrated field must have:

- Source field
- Destination entity
- Destination field
- Transformation rule
- Required or optional status
- Validation rule
- Exception behavior
- Data owner
- Approval status

6.4 Migration Exceptions

The system must produce an exception report for:

- Duplicate customers
- Duplicate projects
- Invalid project codes
- Missing customer references
- Missing locations
- Invalid dates

Invalid financial amounts
Unsupported statuses
Duplicate serial numbers
Missing vendors
Missing document references
Conflicting assignments
Unresolved historical versions
Release-blocking migration exceptions must be resolved before go-live.

6.5 Migration Reconciliation

Reconcile at minimum:

Customer counts
Project counts
Project-code counts
Active and cancelled projects
Document counts
Payment totals
Receivable totals
Payable totals
Inventory quantities
Inventory values
Fixed-asset totals
Loan balances
Contractor records
Audit or historical-reference counts, where available

7. Deployment and Migration Controls

Before production migration:

Confirm Sprint 15 acceptance.
Confirm no release-blocking defects remain.
Create a complete backup.
Freeze source-data changes according to the migration plan.
Confirm migration version.
Confirm rollback point.
Confirm migration approvers.
Confirm support availability.
Confirm communications plan.
During migration:

Record start and end times.
Record migration version.
Record source snapshot.
Record record counts.
Record rejected records.
Record transformed records.
Record warnings.
Record operator identity.
Record audit events.
After migration:

Run reconciliation.
Run smoke tests.

Confirm permissions.
Confirm document access.
Confirm report generation.
Confirm audit-chain integrity.
Confirm backups.
Obtain migration approval.
8. Backup and Recovery Operations
8.1 Backup Procedures
Backups must cover:

Database
File metadata
Stored files
Templates
Artwork
Configuration
Audit logs
Notification and email history
Search-index rebuild information
Backup procedures must document:

Schedule
Backup type
Storage destination
Encryption
Retention
Integrity verification
Restoration owner
Failure notification
Backup-test frequency
8.2 Restore Procedures
The restore runbook must document:

Incident classification
Restore authorization
Recovery environment selection
Backup selection
Database restoration
File restoration
Configuration restoration
Search-index rebuild
Audit-chain verification
Ledger verification
Application smoke testing
Business validation
Recovery approval
Return to service
8.3 Recovery Objectives
Recovery Point Objective and Recovery Time Objective must be documented before go-live.

If final values are not yet approved, the system must record them as operational decisions rather than silently assuming them.

8.4 Restore Testing

Restore testing must verify:

- Record counts
- Foreign-key relationships
- Financial balances
- File references
- File checksums
- Audit-chain integrity
- User and role configuration
- Report reproducibility
- Search functionality
- Notification configuration

9. Monitoring and Alerting

9.1 System Monitoring

Monitor:

- Application availability
- Database availability
- Database storage
- File-storage capacity
- Background-job queues
- Email queues
- Signature-service connectivity
- Search-service availability
- Backup completion
- Restore-test status
- Error rates
- Authentication failures
- Permission failures
- Audit-chain verification
- Report-generation failures

9.2 Business Monitoring

Monitor:

- Failed payment verification
- Failed email delivery
- Expired signature requests
- Overdue tasks
- Unpaid invoices
- Unverified payments
- Payables past due
- Commission-report failures
- Financial-report failures
- Missing required documents
- Data-integrity exceptions
- Negative-profit projects
- Locked-period adjustment requests

9.3 Alert Severity

Permitted alert severities:

INFO

WARNING

` HIGH
CRITICAL

Critical alerts include:

Database unavailable
File storage unavailable
Backup failure
Restore failure
Audit-chain failure
Unbalanced ledger
Unauthorized access pattern
Data corruption
Failed production migration

9.4 Alert Handling

Each operational alert must have:

Alert ID
Severity
Source
Date and time
Description
Affected service or record
Assigned owner
Status
Resolution
Resolution date
Related incident
Audit reference

10. Operational Runbooks

The following runbooks must be delivered.

10.1 User Administration

Create user
Disable user
Change role
Remove access
Reset authentication
Review access
Process contractor departure

10.2 Data Administration

Correct authorized data
Resolve duplicate customer
Resolve duplicate project
Correct serial number
Correct project assignment
Restore superseded record
Process formal adjustment

10.3 Document Administration

Upload artwork
Retire artwork
Upload approved template
Replace a template version
Restore archived file

Resolve file-processing failure
Resolve document-generation failure

10.4 Financial Administration

Open reporting period
Prepare reports
Process report approval
Publish reports
Lock period
Create formal adjustment
Reconcile ledger
Resolve payment discrepancy

10.5 Backup and Recovery

Confirm backup
Investigate backup failure
Restore database
Restore files
Verify audit chain
Verify financial balances
Return service to production

10.6 Incident Response

Identify incident
Classify severity
Preserve evidence
Restrict affected access
Notify responsible roles
Correct or contain issue
Validate recovery
Document root cause
Close incident

11. User and Administrator Documentation

11.1 User Guides

Provide role-specific instructions for:

Login
Dashboard
Search
Customer records
Project records
Master Sales Record
Documents
Signatures
Payments
Installation tasks
Payables
Commissions
Financial reports
Notifications
File access
Error handling

11.2 Administrator Guide

The administrator guide must cover:

User setup

Role assignment
Configuration
Templates
Artwork
Email settings
Signature settings
Backups
Restore procedures
Audit verification
Data corrections
Monitoring
Incident response
Release procedures
Support escalation
11.3 Financial Operations Guide
Provide instructions for:

Period preparation
Report review
Report approval
Report signature
Publication
Locking
Formal adjustments
Reconciliation
Financial-report distribution
12. Release and Rollback
12.1 Release Checklist
Before release, confirm:

Sprint 15 approval
Release candidate identified
Database migration scripts tested
Backup completed
Configuration approved
User list approved
Role assignments approved
Templates approved
Artwork approved
Email configuration tested
Signature configuration tested
Migration data approved
Monitoring active
Support coverage arranged
Rollback decision authority assigned
Go-live communication prepared
12.2 Rollback Conditions
Rollback must be considered when:

Migration corrupts data
Financial balances fail reconciliation
Audit-chain integrity fails
Critical permissions are incorrect

Core workflow is unavailable
Required documents cannot be generated
Backup or restore validation fails
Critical integration services are unavailable
Data loss is detected
12.3 Rollback Controls
Rollback must:

Be authorized.
Preserve logs and evidence.
Use the approved rollback point.
Preserve the failed release for investigation.
Prevent duplicate migration.
Verify restored data.
Reconcile financial records.
Communicate system status.
Record the decision and reason.
13. Go-Live Procedure
The go-live procedure must include:

Final release approval.
Production backup.
Source-data freeze, if applicable.
Production deployment.
Database migration.
Data migration.
Configuration validation.
User and role validation.
Smoke tests.
Financial reconciliation.
Document-generation test.
Email and signature test.
Backup verification.
Monitoring verification.
Business-owner approval.
Go-live communication.
Post-deployment monitoring.
14. Post-Deployment Validation
Within the approved post-deployment period, verify:

Users can log in.
Roles are correct.
Dashboard loads.
Search returns authorized results.
Customers and projects are available.
Project codes are preserved.
Documents open correctly.
New documents generate correctly.
Email delivery works.
Signature requests work.
Payment entry and verification work.
Payables are visible.
Commission records calculate correctly.

Financial reports generate.
Audit entries are created.
Backups complete.
Monitoring alerts function.
No release-blocking defect has appeared.
Post-deployment findings must be recorded and classified as:

Defect
Configuration correction
Training issue
Data issue
Enhancement
Operational incident
15. Operational Ownership
Operational ownership must be assigned for:

Application administration
Database administration
User administration
Financial reporting
Document templates
License artwork
Email delivery
Electronic signatures
File management
Backups
Recovery
Security incidents
Audit verification
Vendor support
Release approval
Each owner must have:

Name or role
Responsibilities
Escalation path
Backup owner
Availability expectations
16. Sprint 16 Decisions
Decision 16-01 — Production authority
The production relational database is the authoritative system of record.

Decision 16-02 — Controlled deployment
Production deployments require an approved release version, deployment checklist, backup, and rollback plan.

Decision 16-03 — Separate environments
Development, test, user acceptance, and production environments remain separate after go-live.

Decision 16-04 — Migration approval
Production migration requires reconciliation evidence and approval by authorized project and financial owners.

Decision 16-05 — Backup before change

A verified backup must be created before production deployment, database migration, configuration changes affecting financial records, or destructive administrative operations.

Decision 16-06 — No unapproved production changes

Production changes must be documented, authorized, and auditable.

Decision 16-07 — Least-privilege administration

Administrative access is limited to the minimum necessary users and is reviewed periodically.

Decision 16-08 — Secrets management

Production credentials, tokens, encryption keys, and signature-related secrets must not be stored in source code or ordinary user documentation.

Decision 16-09 — Rollback authority

A designated release owner may initiate rollback when a release-blocking defect or production integrity issue is confirmed.

Decision 16-10 — Financial rollback caution

A release rollback must not silently reverse valid business transactions. Financial corrections require controlled reconciliation and, when necessary, formal adjustments.

Decision 16-11 — Monitoring

Production monitoring covers technical availability, integration failures, data integrity, audit integrity, financial reconciliation, and operational exceptions.

Decision 16-12 — Operational documentation

No production handover is complete until administrator guides, user guides, runbooks, and escalation procedures are delivered.

Decision 16-13 — Post-deployment review

The release requires a documented post-deployment validation review before it is considered fully accepted.

Decision 16-14 — Support ownership

Operational owners and backup owners must be assigned before go-live.

Decision 16-15 — Historical preservation

Migration and deployment activities must preserve approved historical records, document versions, audit entries, signatures, financial records, and source references.

17. Open Questions

Open Question 16-01 — Hosting model

Will the production system be hosted:

In a managed cloud environment

On company-controlled infrastructure

Through a third-party application platform

Recommended default: Use managed infrastructure with documented backups, access controls, monitoring, and recovery capabilities.

Open Question 16-02 — Recovery objectives

What are the approved:

Recovery Point Objective

Recovery Time Objective

Recommended default: Approve these before production deployment and test restoration against them.

Open Question 16-03 — Backup schedule

What backup schedule is required?

Recommended default: Daily full or equivalent backups, more frequent incremental protection where supported, and periodic restore testing.

Open Question 16-04 — Operational support hours

What support hours apply after go-live?

Recommended default: Define normal business-hour support with an escalation path for critical production incidents.

Open Question 16-05 — Go-live date

What date and time are approved for production cutover?

Recommended default: Use a low-activity period with confirmed support availability and a defined rollback window.

Open Question 16-06 — Migration freeze

How long must source systems be frozen during migration?

Recommended default: Freeze source changes during the final extraction, migration, reconciliation, and acceptance window.

Open Question 16-07 — Initial migration scope

Will all historical records be migrated, or only active and required historical records?

Recommended default: Migrate all records required for legal, financial, audit, customer, and project continuity. Document any intentionally excluded records.

Open Question 16-08 — Production administrator

Who is the primary Database Administrator after go-live?

Recommended default: Assign one primary administrator and one backup administrator before deployment.

Open Question 16-09 — Monitoring ownership

Who receives and resolves critical technical and business alerts?

Recommended default: Assign technical monitoring to the system administrator and financial/data alerts to the Comptroller or designated owner.

Open Question 16-10 — Configuration approval

Who approves changes to:

Commission rates

Default costs

Artwork

Document templates

Report formulas

Notification templates

Permission rules

Recommended default: Require role-specific approval and an audit record for each configuration class.

Open Question 16-11 — Incident notification

Which incidents require notification to users, contractors, customers, vendors, or external authorities?

Recommended default: Define notification rules during operational handover and require documented incident classification.

Open Question 16-12 — Maintenance windows

When may planned maintenance occur?

Recommended default: Schedule maintenance outside normal operating hours and notify affected users in advance.

Open Question 16-13 — Post-go-live stabilization period

How long will enhanced support remain active after deployment?

Recommended default: Use a defined stabilization period with daily issue review and formal handoff to normal operations.

Open Question 16-14 — Future bundle after Sprint 16

The approved baseline ends at Sprint 16. Should a subsequent bundle be created for ongoing operations and enhancements?

Recommended default: Create GECC-17-post-deployment-operations.md after go-live to document stabilization findings, approved enhancements, recurring maintenance, and operational metrics.

18. Acceptance Criteria

Sprint 16 is accepted when:

Production deployment instructions are complete.

Production configuration is documented and approved.

Secrets are protected using an approved mechanism.

Database migration scripts are tested.

File-storage configuration is tested.

Email configuration is tested.

Electronic-signature configuration is tested.

Search and background-job services are tested.

Initial users are created with approved roles.

Role and permission validation is complete.

Migration mapping is documented.

Test migration is complete.

Migration reconciliation is approved.

Production migration is completed with an audit record.

Customer, project, project-code, document, payment, payable, inventory, asset, loan, and contractor records reconcile.

Pre-deployment backups are completed and verified.

Restore procedures are documented and tested.

Monitoring and alerting are operational.

Audit-chain verification is available to authorized users.

Backup failures and critical system failures generate alerts.

Administrator documentation is complete.

Role-specific user guides are complete.

Financial operations documentation is complete.

Operational runbooks are complete.

Rollback procedures are documented and tested to the approved extent.

Go-live smoke tests pass.

Post-deployment validation passes.

No unresolved Blocker or Critical defects remain.

Deferred High-severity defects have documented approval and owners.

Operational ownership and backup ownership are assigned.

Support and escalation procedures are active.

Historical records and versions remain preserved.

The production system remains auditable after migration and deployment.

Sprint 16 go-live approval is documented.

19. Deployment Test Scenarios

Environment Tests

Deploy the approved release to a clean environment.

Apply database migrations.

Verify application and database versions.

Verify configuration values.

Verify secrets are not exposed.

Verify background jobs.

Verify email service.

Verify signature service.

Verify file storage.

Verify search service.

Migration Tests

Run a test migration.

Reconcile customer counts.

Reconcile project counts.

Reconcile project codes.

Reconcile documents.

Reconcile payment and payable totals.

Reconcile inventory and asset records.

Reconcile loans and contractors.

Resolve migration exceptions.

Repeat migration from the approved source snapshot.

Production Smoke Tests

Log in as each initial role.

Open the dashboard.

Search for an authorized project.

Open a customer.

Open a project.

Generate a test document.

Verify file storage.

Send a test email.

Create a test notification.
Verify audit logging.
Run a financial report.
Verify ledger balance.
Confirm restricted actions are blocked.
Recovery Tests
Restore the production backup into the recovery environment.
Verify record counts.
Verify file references.
Verify checksums.
Verify audit-chain integrity.
Verify ledger balances.
Reproduce a published financial report.
Rebuild search indexes.
Confirm role and permission configuration.
20. Sprint 16 Completion Definition
Sprint 16 is complete when:

The production deployment package is approved.
Production configuration is documented.
Initial users and roles are configured.
Data migration is complete and reconciled.
Backup and restore procedures are operational.
Monitoring and alerting are active.
Administrator, user, and financial guides are delivered.
Runbooks and escalation procedures are delivered.
Rollback procedures are documented.
Go-live smoke testing passes.
Post-deployment validation passes.
Operational ownership is assigned.
No release-blocking defects remain.
The production system preserves historical records and audit integrity.
Sprint 16 acceptance is approved.
21. Next Markdown Bundle
The approved baseline ends with Sprint 16. The recommended next bundle is:

text

GECC-17-post-deployment-operations.md
Its objective would be to document:

Go-live stabilization findings
Production incidents
Operational metrics
Backup and restore results
User feedback
Approved enhancements
Deferred defects
Performance improvements
Permission reviews
Data-quality reviews
Recurring maintenance
Configuration changes

Audit-log growth and archival

Disaster-recovery exercises

New business requirements

Next-release planning

Lessons learned

Operational acceptance

Sprint 17 should be created only after Sprint 16 go-live validation produces operational findings or approved enhancements.

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